SCM Code Outputs:

Problem: p5

497_561_564:

```
Command Prompt - python

C:\Users\rockn\OneDrive\Desktop\SCM>python -W ignore p5_scm_arima_forecast.py
==============================================================
  P5 SUPPLYNEX — ARIMA DEMAND FORECASTING
  Course: UE23CS342BA1  |  Team: SupplyNex
==============================================================

──────────────────────────────────────────────────────────────
  Product : Laptop Pro X (Electronics)
──────────────────────────────────────────────────────────────

  [ADF Stationarity Test on Training Data]
  ADF Statistic : 1.3602
  p-value       : 0.9969
  Stationary    : NO — differencing needed

  [Grid Search — Best ARIMA(p,d,q)]
C:\Users\rockn\AppData\Roaming\Python\Python312\site-packages\statsmodels\base\model.py:607: ConvergenceWarning: Maximum Likelihood optimization failed to converge. Check mle_retvals
  warnings.warn("Maximum Likelihood optimization failed to "
C:\Users\rockn\AppData\Roaming\Python\Python312\site-packages\statsmodels\base\model.py:607: ConvergenceWarning: Maximum Likelihood optimization failed to converge. Check mle_retvals
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  warnings.warn("Maximum Likelihood optimization failed to "
  Best order : ARIMA(2, 2, 0)  |  AIC = 77.86

  [Validation — 3 months held out]
  MAE  = 11.80 units
  RMSE = 12.52 units
  MAPE = 5.33%

  [6-Month Demand Forecast]
  May 2026 → 230 units
  Jun 2026 → 235 units
  Jul 2026 → 240 units
  Aug 2026 → 245 units
```

```
Command Prompt - python

  [6-Month Demand Forecast]
  May 2026 → 230 units
  Jun 2026 → 235 units
  Jul 2026 → 240 units
  Aug 2026 → 245 units
  Sep 2026 → 250 units
  Oct 2026 → 255 units

──────────────────────────────────────────────────────────────
  Product : Steel Sheets (Raw Materials)
──────────────────────────────────────────────────────────────

  [ADF Stationarity Test on Training Data]
  ADF Statistic : 6.3051
  p-value       : 1.0000
  Stationary    : NO — differencing needed

  [Grid Search — Best ARIMA(p,d,q)]
C:\Users\rockn\AppData\Roaming\Python\Python312\site-packages\statsmodels\base\model.py:607: ConvergenceWarning: Maximum Likelihood optimization failed to converge. Check mle_retvals
  warnings.warn("Maximum Likelihood optimization failed to "
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C:\Users\rockn\AppData\Roaming\Python\Python312\site-packages\statsmodels\base\model.py:607: ConvergenceWarning: Maximum Likelihood optimization failed to converge. Check mle_retvals
  warnings.warn("Maximum Likelihood optimization failed to "
  Best order : ARIMA(2, 2, 1)  |  AIC = 82.90

  [Validation — 3 months held out]
  MAE  = 14.86 units
  RMSE = 15.03 units
  MAPE = 3.77%

  [6-Month Demand Forecast]
  May 2026 → 401 units
  Jun 2026 → 410 units
  Jul 2026 → 415 units
  Aug 2026 → 418 units
```

```
Command Prompt - python -

  [6-Month Demand Forecast]
  May 2026 → 401 units
  Jun 2026 → 410 units
  Jul 2026 → 415 units
  Aug 2026 → 418 units
  Sep 2026 → 425 units
  Oct 2026 → 431 units

  ────────────────────────────────────────────────────────────
  Product : Copper Wire (Raw Materials)
  ────────────────────────────────────────────────────────────

  [ADF Stationarity Test on Training Data]
  ADF Statistic : 0.8786
  p-value       : 0.9928
  Stationary    : NO — differencing needed

  [Grid Search — Best ARIMA(p,d,q)]
C:\Users\rockn\AppData\Roaming\Python\Python312\site-packages\statsmodels\base\model.py:607: ConvergenceWarning: Maximum Likelihood optimization failed to converge. Check mle_retvals
  warnings.warn("Maximum Likelihood optimization failed to "
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C:\Users\rockn\AppData\Roaming\Python\Python312\site-packages\statsmodels\base\model.py:607: ConvergenceWarning: Maximum Likelihood optimization failed to converge. Check mle_retvals
  warnings.warn("Maximum Likelihood optimization failed to "
  Best order : ARIMA(0, 2, 2)  |  AIC = 80.25
```

```
Command Prompt - python -

  Best order : ARIMA(0, 2, 2)  |  AIC = 80.25

  [Validation — 3 months held out]
  MAE  = 6.51 units
  RMSE = 8.27 units
  MAPE = 2.58%

  [6-Month Demand Forecast]
  May 2026 → 264 units
  Jun 2026 → 268 units
  Jul 2026 → 273 units
  Aug 2026 → 277 units
  Sep 2026 → 281 units
  Oct 2026 → 286 units

  ────────────────────────────────────────────────────────────
  Product : Rice 5kg Pack (FMCG)
  ────────────────────────────────────────────────────────────

  [ADF Stationarity Test on Training Data]
  ADF Statistic : 1.4403
  p-value       : 0.9973
  Stationary    : NO — differencing needed

  [Grid Search — Best ARIMA(p,d,q)]
C:\Users\rockn\AppData\Roaming\Python\Python312\site-packages\statsmodels\base\model.py:607: ConvergenceWarning: Maximum Likelihood optimization failed to converge. Check mle_retvals
  warnings.warn("Maximum Likelihood optimization failed to "
C:\Users\rockn\AppData\Roaming\Python\Python312\site-packages\statsmodels\base\model.py:607: ConvergenceWarning: Maximum Likelihood optimization failed to converge. Check mle_retvals
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C:\Users\rockn\AppData\Roaming\Python\Python312\site-packages\statsmodels\base\model.py:607: ConvergenceWarning: Maximum Likelihood optimization failed to converge. Check mle_retvals
  warnings.warn("Maximum Likelihood optimization failed to "
  Best order : ARIMA(0, 2, 2)  |  AIC = 95.32

  [Validation — 3 months held out]
  MAE  = 7.60 units
  RMSE = 7.71 units
```

```
Command Prompt - python -

  [Validation — 3 months held out]
  MAE  = 7.60 units
  RMSE = 7.71 units
  MAPE = 1.25%

  [6-Month Demand Forecast]
  May 2026 → 629 units
  Jun 2026 → 638 units
  Jul 2026 → 647 units
  Aug 2026 → 656 units
  Sep 2026 → 666 units
  Oct 2026 → 675 units

────────────────────────────────────────────────────────────
  Product : Cardboard Box (Packaging)
────────────────────────────────────────────────────────────

  [ADF Stationarity Test on Training Data]
  ADF Statistic : 2.8332
  p-value       : 1.0000
  Stationary    : NO — differencing needed

  [Grid Search — Best ARIMA(p,d,q)]
C:\Users\rockn\AppData\Roaming\Python\Python312\site-packages\statsmodels\base\model.py:607: ConvergenceWarning: Maximum Likelihood optimization failed to converge. Check mle_retvals
  warnings.warn("Maximum Likelihood optimization failed to "
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C:\Users\rockn\AppData\Roaming\Python\Python312\site-packages\statsmodels\base\model.py:607: ConvergenceWarning: Maximum Likelihood optimization failed to converge. Check mle_retvals
  warnings.warn("Maximum Likelihood optimization failed to "
  Best order : ARIMA(0, 2, 3)  |  AIC = 77.11
```

```
Command Prompt - python -

C:\Users\rockn\AppData\Roaming\Python\Python312\site-packages\statsmodels\base\model.py:607: ConvergenceWarning: Maximum Likelihood optimization failed to converge. Check mle_retvals
  warnings.warn("Maximum Likelihood optimization failed to "
  Best order : ARIMA(0, 2, 3)  |  AIC = 77.11

  [Validation — 3 months held out]
  MAE  = 5.20 units
  RMSE = 6.41 units
  MAPE = 2.59%

  [6-Month Demand Forecast]
  May 2026 → 209 units
  Jun 2026 → 212 units
  Jul 2026 → 216 units
  Aug 2026 → 220 units
  Sep 2026 → 223 units
  Oct 2026 → 227 units


============================================================
  EXPORT COMPLETE → p5_arima_forecast_output.csv
============================================================
                product_name forecast_month  forecasted_units     model_used  mape_pct
  Laptop Pro X (Electronics)     2026-05-01               230 ARIMA(2, 2, 0)      5.33
  Laptop Pro X (Electronics)     2026-06-01               235 ARIMA(2, 2, 0)      5.33
  Laptop Pro X (Electronics)     2026-07-01               240 ARIMA(2, 2, 0)      5.33
  Laptop Pro X (Electronics)     2026-08-01               245 ARIMA(2, 2, 0)      5.33
  Laptop Pro X (Electronics)     2026-09-01               250 ARIMA(2, 2, 0)      5.33
  Laptop Pro X (Electronics)     2026-10-01               255 ARIMA(2, 2, 0)      5.33
Steel Sheets (Raw Materials)     2026-05-01               401 ARIMA(2, 2, 1)      3.77
Steel Sheets (Raw Materials)     2026-06-01               410 ARIMA(2, 2, 1)      3.77
Steel Sheets (Raw Materials)     2026-07-01               415 ARIMA(2, 2, 1)      3.77
Steel Sheets (Raw Materials)     2026-08-01               418 ARIMA(2, 2, 1)      3.77
Steel Sheets (Raw Materials)     2026-09-01               425 ARIMA(2, 2, 1)      3.77
Steel Sheets (Raw Materials)     2026-10-01               431 ARIMA(2, 2, 1)      3.77
 Copper Wire (Raw Materials)     2026-05-01               264 ARIMA(0, 2, 2)      2.58
 Copper Wire (Raw Materials)     2026-06-01               268 ARIMA(0, 2, 2)      2.58
 Copper Wire (Raw Materials)     2026-07-01               273 ARIMA(0, 2, 2)      2.58
 Copper Wire (Raw Materials)     2026-08-01               277 ARIMA(0, 2, 2)      2.58
 Copper Wire (Raw Materials)     2026-09-01               281 ARIMA(0, 2, 2)      2.58
 Copper Wire (Raw Materials)     2026-10-01               286 ARIMA(0, 2, 2)      2.58
        Rice 5kg Pack (FMCG)     2026-05-01               629 ARIMA(0, 2, 2)      1.25
        Rice 5kg Pack (FMCG)     2026-06-01               638 ARIMA(0, 2, 2)      1.25
        Rice 5kg Pack (FMCG)     2026-07-01               647 ARIMA(0, 2, 2)      1.25
        Rice 5kg Pack (FMCG)     2026-08-01               656 ARIMA(0, 2, 2)      1.25
        Rice 5kg Pack (FMCG)     2026-09-01               666 ARIMA(0, 2, 2)      1.25
        Rice 5kg Pack (FMCG)     2026-10-01               675 ARIMA(0, 2, 2)      1.25
   Cardboard Box (Packaging)     2026-05-01               209 ARIMA(0, 2, 3)      2.59
   Cardboard Box (Packaging)     2026-06-01               212 ARIMA(0, 2, 3)      2.59
   Cardboard Box (Packaging)     2026-07-01               216 ARIMA(0, 2, 3)      2.59
   Cardboard Box (Packaging)     2026-08-01               220 ARIMA(0, 2, 3)      2.59
```

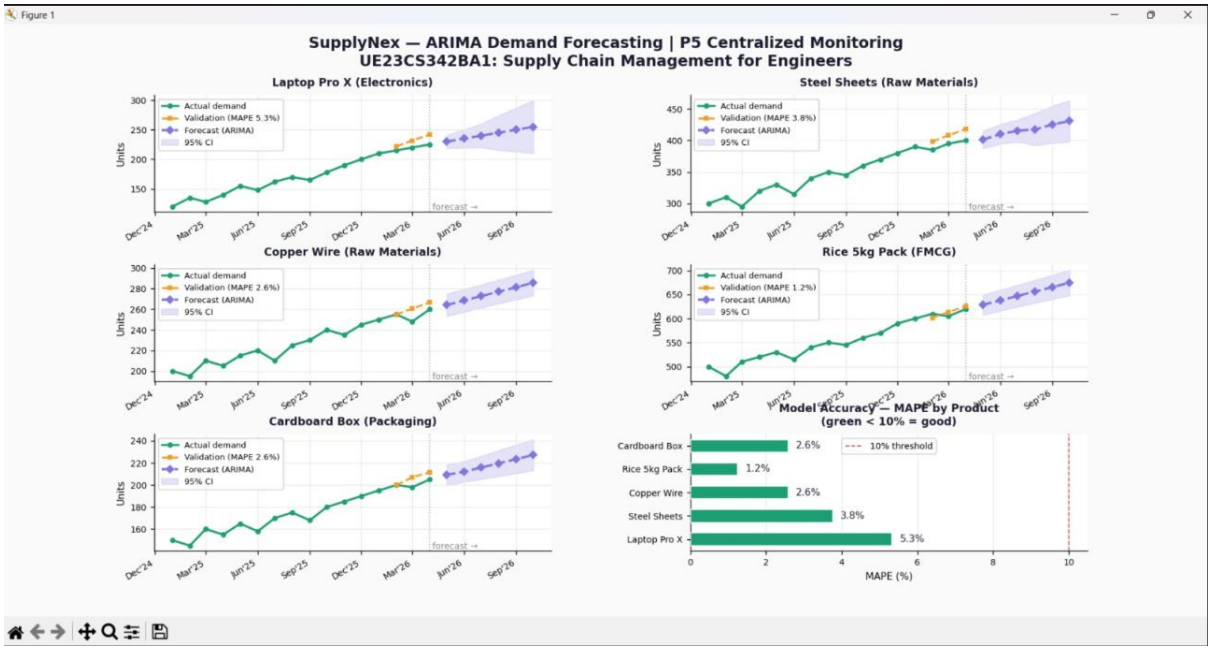
Command Prompt - python

```
Copper Wire (Raw Materials)     2026-09-01              281 ARIMA(0, 2, 2)      2.58
Copper Wire (Raw Materials)     2026-10-01              286 ARIMA(0, 2, 2)      2.58
       Rice 5kg Pack (FMCG)     2026-05-01              629 ARIMA(0, 2, 2)      1.25
       Rice 5kg Pack (FMCG)     2026-06-01              638 ARIMA(0, 2, 2)      1.25
       Rice 5kg Pack (FMCG)     2026-07-01              647 ARIMA(0, 2, 2)      1.25
       Rice 5kg Pack (FMCG)     2026-08-01              656 ARIMA(0, 2, 2)      1.25
       Rice 5kg Pack (FMCG)     2026-09-01              666 ARIMA(0, 2, 2)      1.25
       Rice 5kg Pack (FMCG)     2026-10-01              675 ARIMA(0, 2, 2)      1.25
  Cardboard Box (Packaging)     2026-05-01              209 ARIMA(0, 2, 3)      2.59
  Cardboard Box (Packaging)     2026-06-01              212 ARIMA(0, 2, 3)      2.59
  Cardboard Box (Packaging)     2026-07-01              216 ARIMA(0, 2, 3)      2.59
  Cardboard Box (Packaging)     2026-08-01              220 ARIMA(0, 2, 3)      2.59
  Cardboard Box (Packaging)     2026-09-01              223 ARIMA(0, 2, 3)      2.59
  Cardboard Box (Packaging)     2026-10-01              227 ARIMA(0, 2, 3)      2.59


-- — SQL INSERT for DEMAND_FORECAST table ——————————————
-- Paste these rows after your existing INSERT statements

INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (1, 1, '2026-05-01', 230, NULL, 'ARIMA(2, 2, 0)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (1, 1, '2026-06-01', 235, NULL, 'ARIMA(2, 2, 0)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (1, 1, '2026-07-01', 240, NULL, 'ARIMA(2, 2, 0)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (1, 1, '2026-08-01', 245, NULL, 'ARIMA(2, 2, 0)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (1, 1, '2026-09-01', 250, NULL, 'ARIMA(2, 2, 0)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (1, 1, '2026-10-01', 255, NULL, 'ARIMA(2, 2, 0)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (4, 3, '2026-05-01', 401, NULL, 'ARIMA(2, 2, 1)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (4, 3, '2026-06-01', 410, NULL, 'ARIMA(2, 2, 1)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (4, 3, '2026-07-01', 415, NULL, 'ARIMA(2, 2, 1)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (4, 3, '2026-08-01', 418, NULL, 'ARIMA(2, 2, 1)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (4, 3, '2026-09-01', 425, NULL, 'ARIMA(2, 2, 1)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (4, 3, '2026-10-01', 431, NULL, 'ARIMA(2, 2, 1)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (6, 3, '2026-05-01', 264, NULL, 'ARIMA(0, 2, 2)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (6, 3, '2026-06-01', 268, NULL, 'ARIMA(0, 2, 2)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (6, 3, '2026-07-01', 273, NULL, 'ARIMA(0, 2, 2)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (6, 3, '2026-08-01', 277, NULL, 'ARIMA(0, 2, 2)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (6, 3, '2026-09-01', 281, NULL, 'ARIMA(0, 2, 2)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (6, 3, '2026-10-01', 286, NULL, 'ARIMA(0, 2, 2)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (8, 5, '2026-05-01', 629, NULL, 'ARIMA(0, 2, 2)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (8, 5, '2026-06-01', 638, NULL, 'ARIMA(0, 2, 2)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (8, 5, '2026-07-01', 647, NULL, 'ARIMA(0, 2, 2)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (8, 5, '2026-08-01', 656, NULL, 'ARIMA(0, 2, 2)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (8, 5, '2026-09-01', 666, NULL, 'ARIMA(0, 2, 2)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (8, 5, '2026-10-01', 675, NULL, 'ARIMA(0, 2, 2)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (11, 7, '2026-05-01', 209, NULL, 'ARIMA(0, 2, 3)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (11, 7, '2026-06-01', 212, NULL, 'ARIMA(0, 2, 3)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (11, 7, '2026-07-01', 216, NULL, 'ARIMA(0, 2, 3)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (11, 7, '2026-08-01', 220, NULL, 'ARIMA(0, 2, 3)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (11, 7, '2026-09-01', 223, NULL, 'ARIMA(0, 2, 3)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (11, 7, '2026-10-01', 227, NULL, 'ARIMA(0, 2, 3)');
```

Figure 1




```
-- — SQL INSERT for DEMAND_FORECAST table ——————————————
-- Paste these rows after your existing INSERT statements

INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (1, 1, '2026-05-01', 230, NULL, 'ARIMA(2, 2, 0)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (1, 1, '2026-06-01', 235, NULL, 'ARIMA(2, 2, 0)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (1, 1, '2026-07-01', 240, NULL, 'ARIMA(2, 2, 0)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (1, 1, '2026-08-01', 245, NULL, 'ARIMA(2, 2, 0)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (1, 1, '2026-09-01', 250, NULL, 'ARIMA(2, 2, 0)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (1, 1, '2026-10-01', 255, NULL, 'ARIMA(2, 2, 0)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (4, 3, '2026-05-01', 401, NULL, 'ARIMA(2, 2, 1)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (4, 3, '2026-06-01', 410, NULL, 'ARIMA(2, 2, 1)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (4, 3, '2026-07-01', 415, NULL, 'ARIMA(2, 2, 1)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (4, 3, '2026-08-01', 418, NULL, 'ARIMA(2, 2, 1)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (4, 3, '2026-09-01', 425, NULL, 'ARIMA(2, 2, 1)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (4, 3, '2026-10-01', 431, NULL, 'ARIMA(2, 2, 1)');
INSERT INTO DEMAND_FORECAST (product_id, warehouse_id, forecast_month, forecasted_units, actual_units, model_used) VALUES (6, 3, '2026-05-01', 264, NULL, 'ARIMA(0, 2, 2)');
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  Plot saved → p5_arima_forecast_plots.png
  Done ✓

C:\Users\rockn\OneDrive\Desktop\SCM>
```